Experiment:- 6

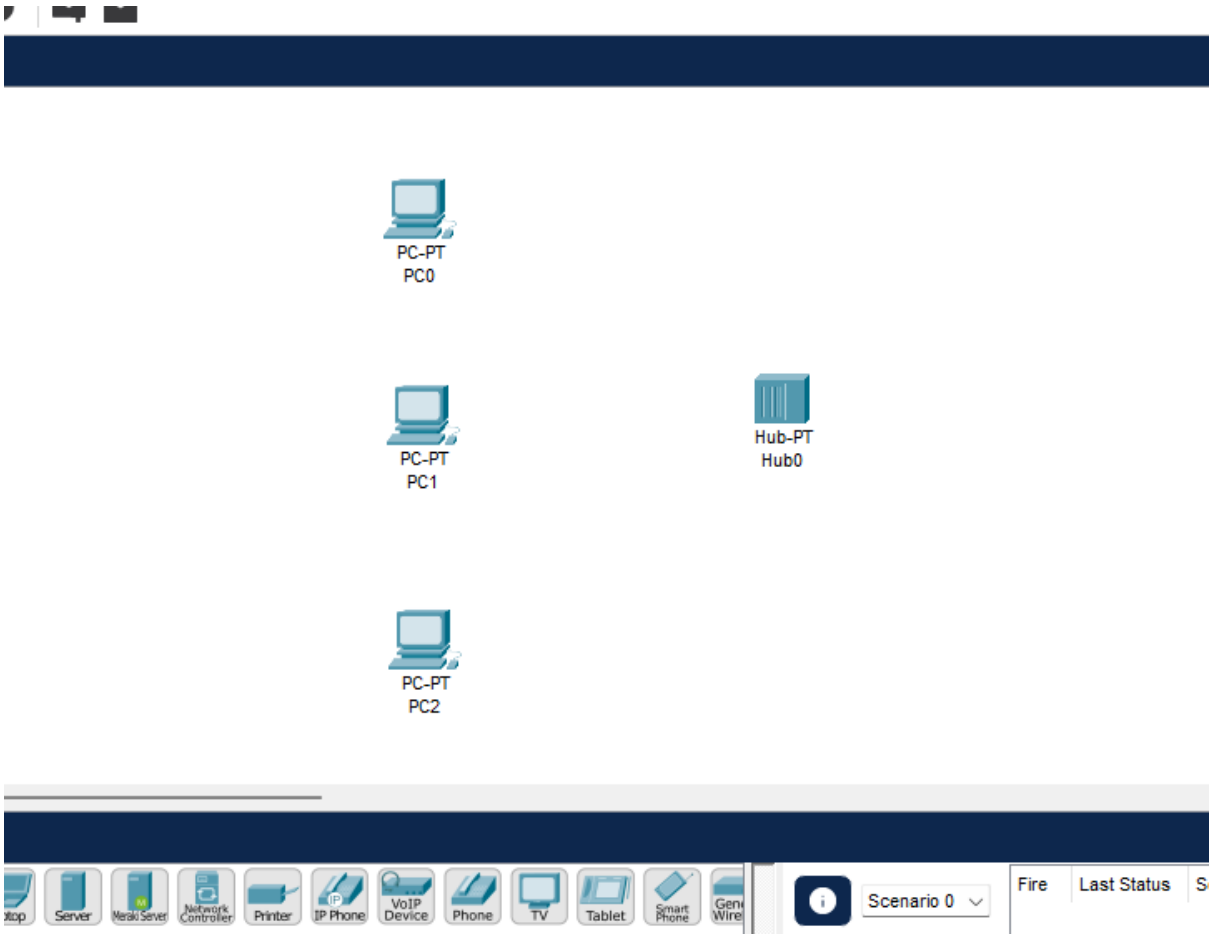
Problem Statement:-

Develop a simulation to demonstrate multiple access protocols such as Pure ALOHA, Slotted ALOHA, CSMA/CD, and CSMA/CA using Cisco Packet Tracer. Analyze the performance of each protocol in handling network collisions and maximizing data transmission efficiency.

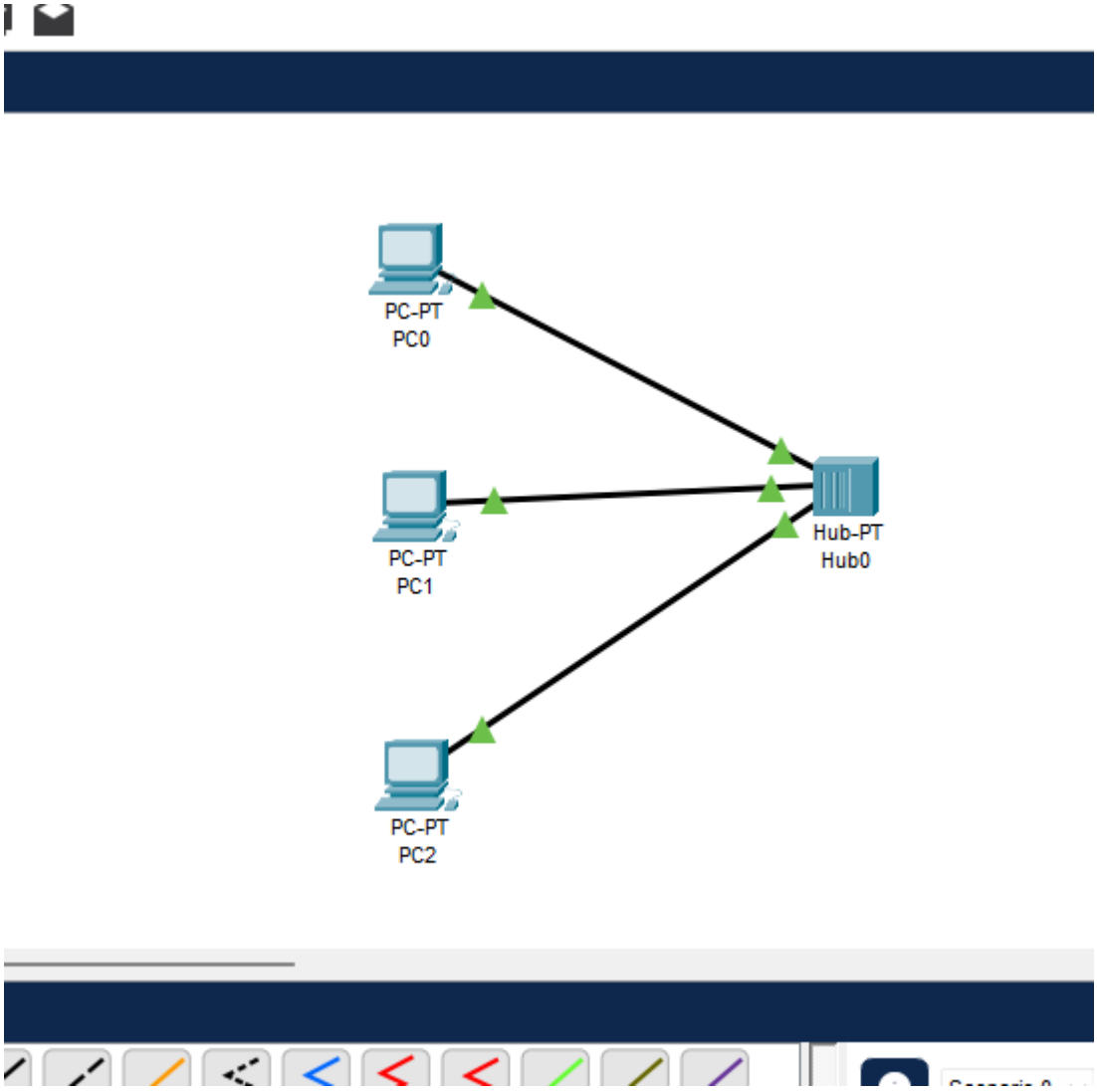
Aim:- **Simulation of Multiple Access Protocols**

PROCEDURE:- **Pure Aloha**

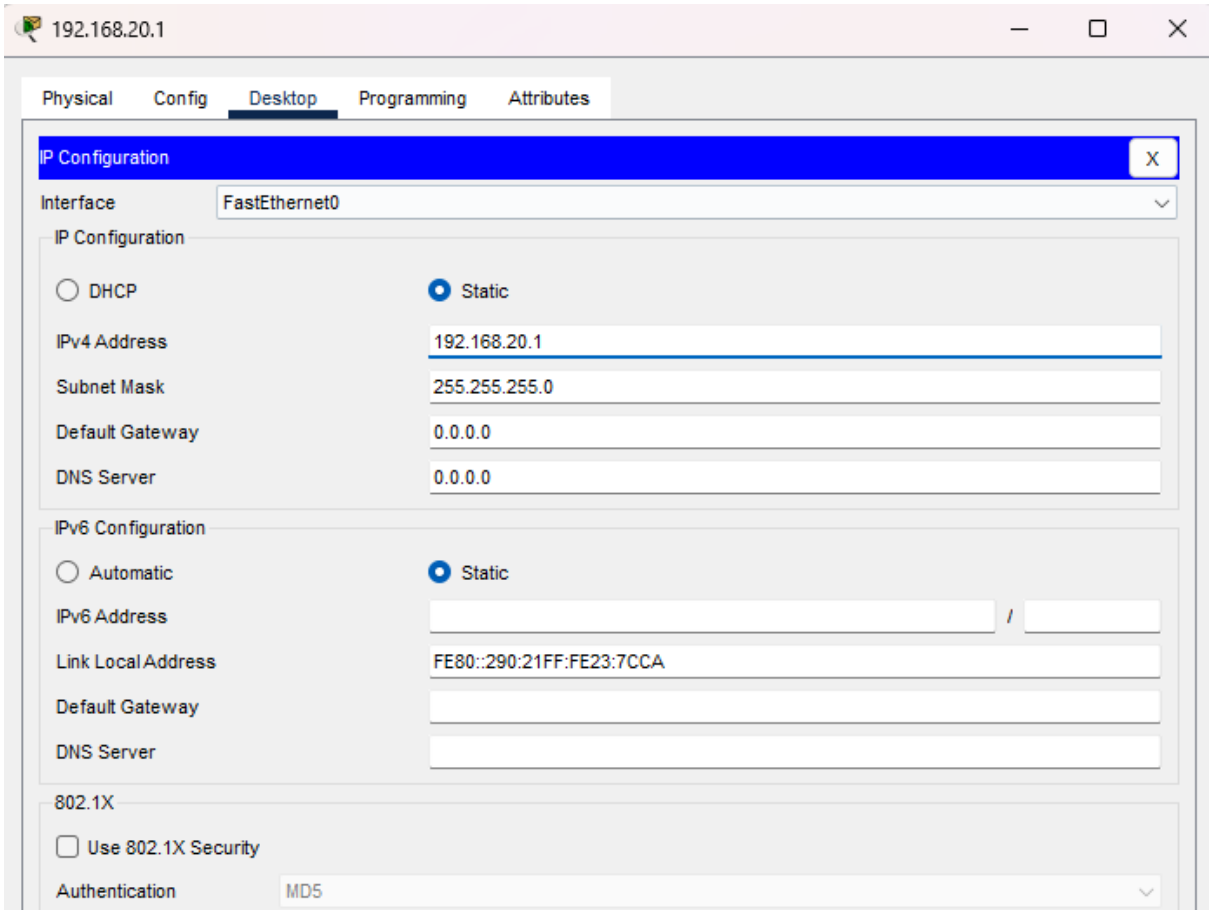
- Open Cisco Packet Tracer and create a new project.
- Add devices:
  - Drag and drop **4–6 PCs**
  - Add **1 Hub-PT**

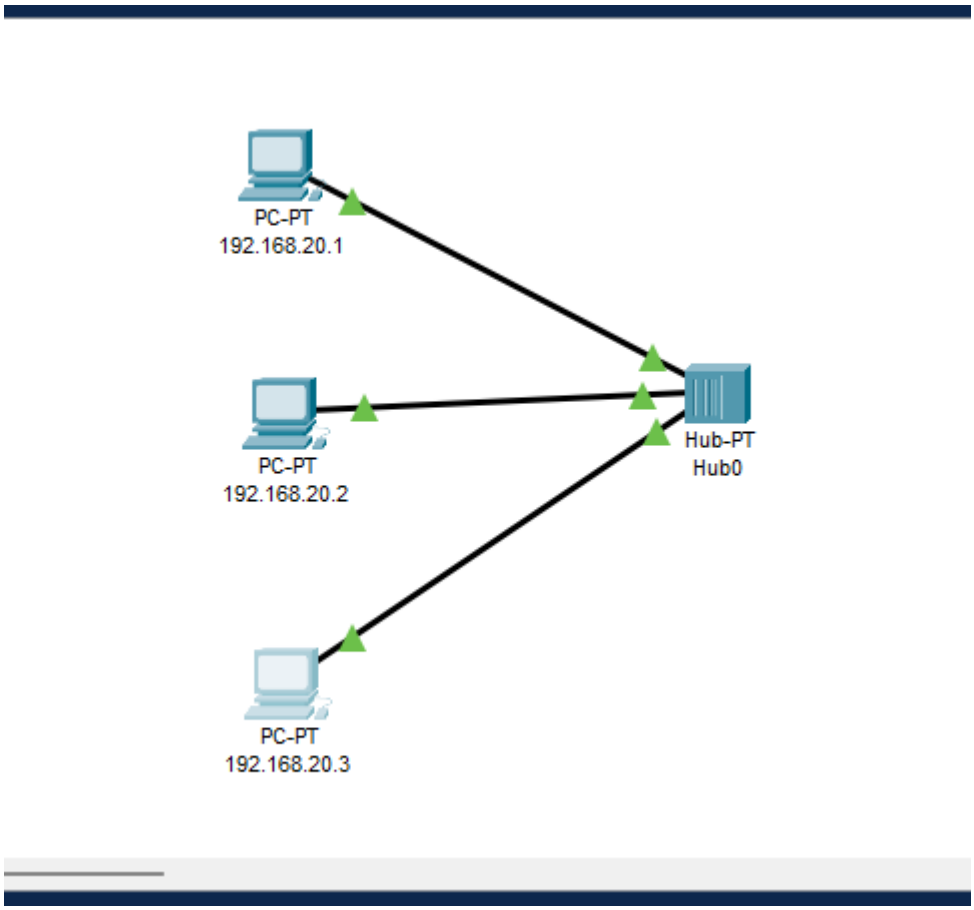


- Connect devices:
  - Use **Copper Straight-Through Cable**
  - Connect all PCs to the hub

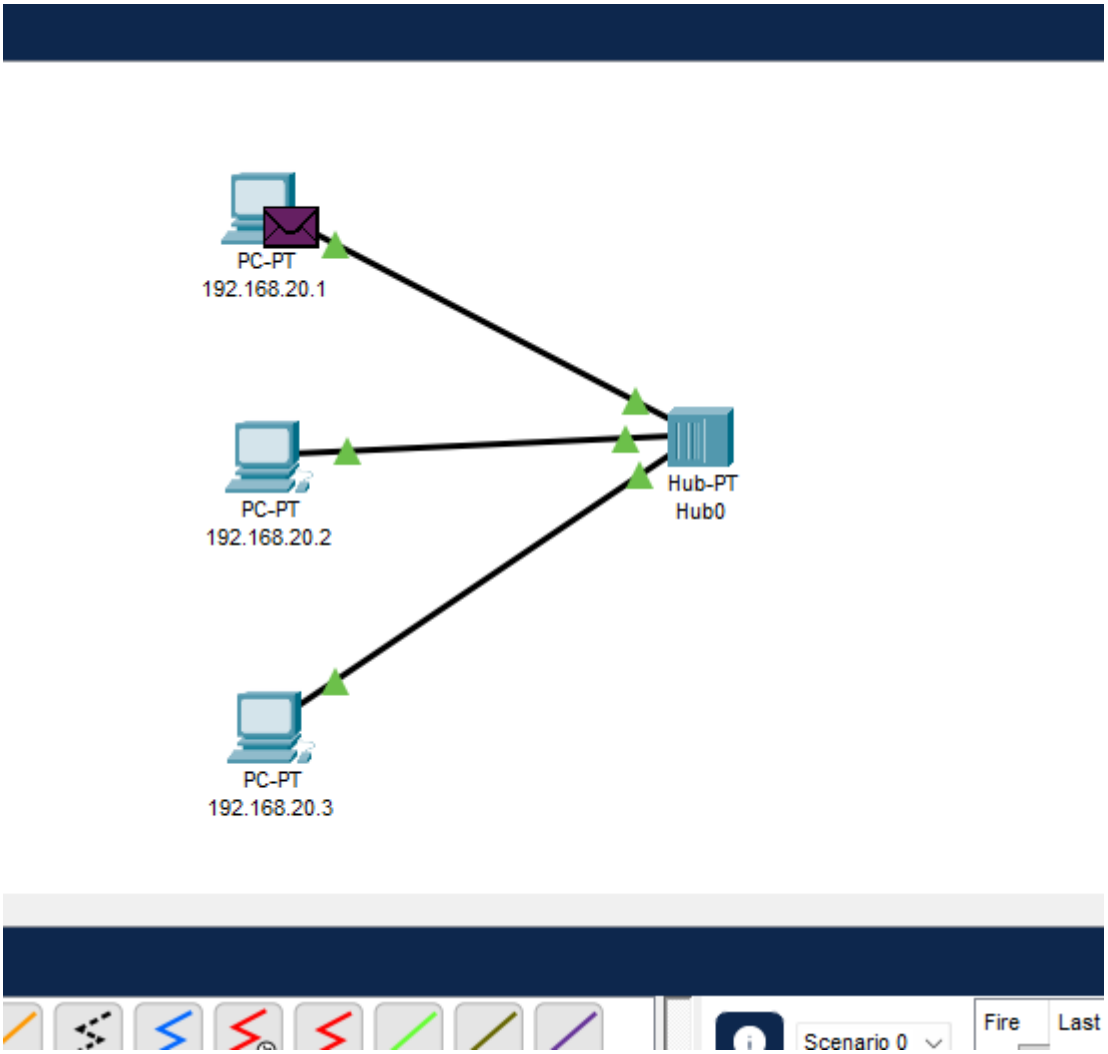


- Configure IP addresses:
    - Assign IP addresses to each PC in the same network

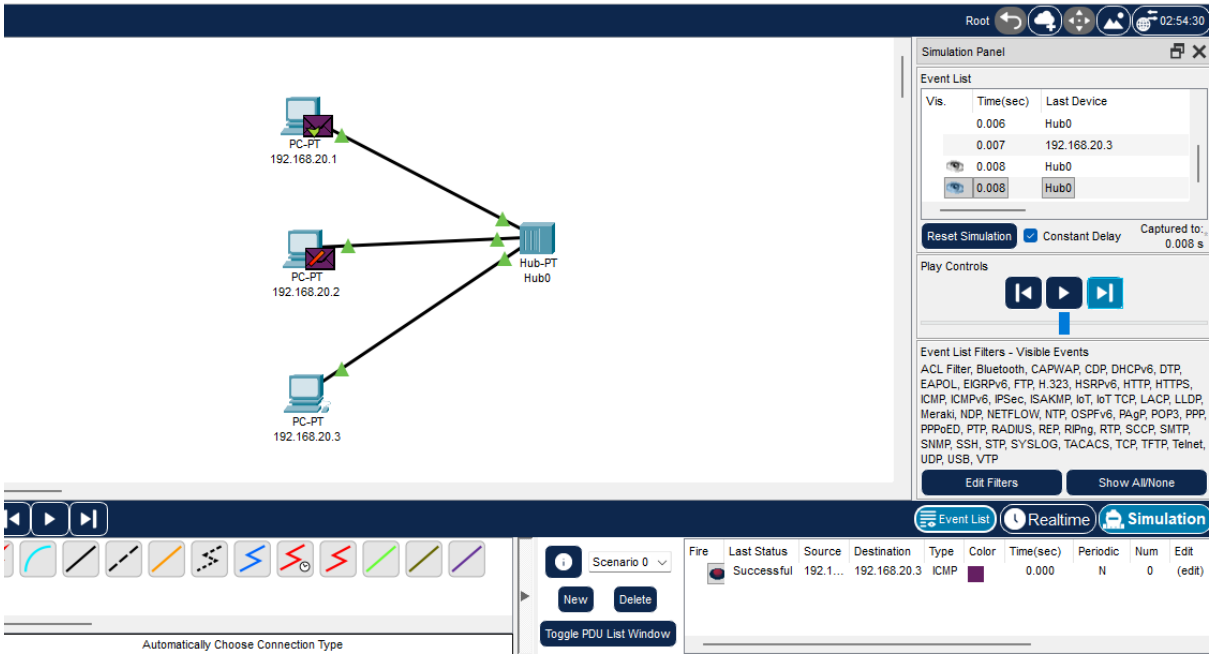

192.168.20.1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.1
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:21FF:FE23:7CCA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5


PC-PT
192.168.20.1
PC-PT
192.168.20.2
PC-PT
192.168.20.3
Hub-PT
Hub0

- Simulate data transmission:
  - Use **Add Simple PDU tool (envelope icon)**
  - Send packets between different PCs



- Enable Simulation Mode:
  - Switch to **Simulation Mode**
  - Click **Auto Capture/Play**



- Observe:

Packet Flow through Hub:
In, the hub broadcasts packets to all devices, but only the destination accepts it.

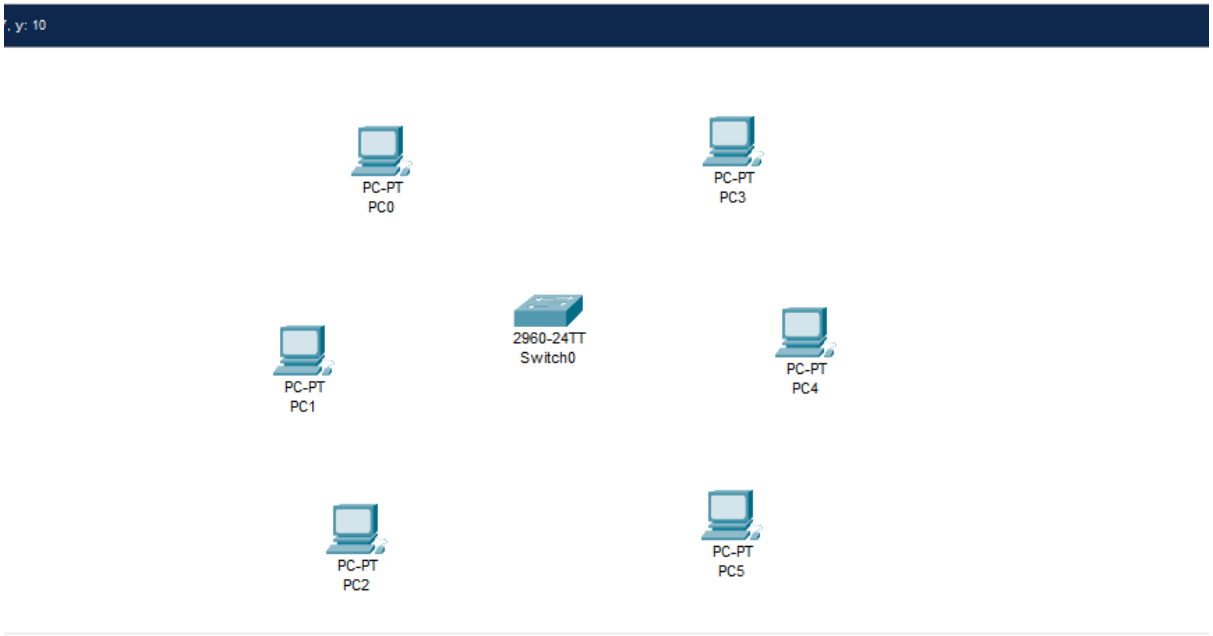
Collisions between Packets:
When multiple devices send data simultaneously, packets overlap, causing collisions.

Retransmissions after Collision:
After collision, data is lost and devices retransmit after a random delay.

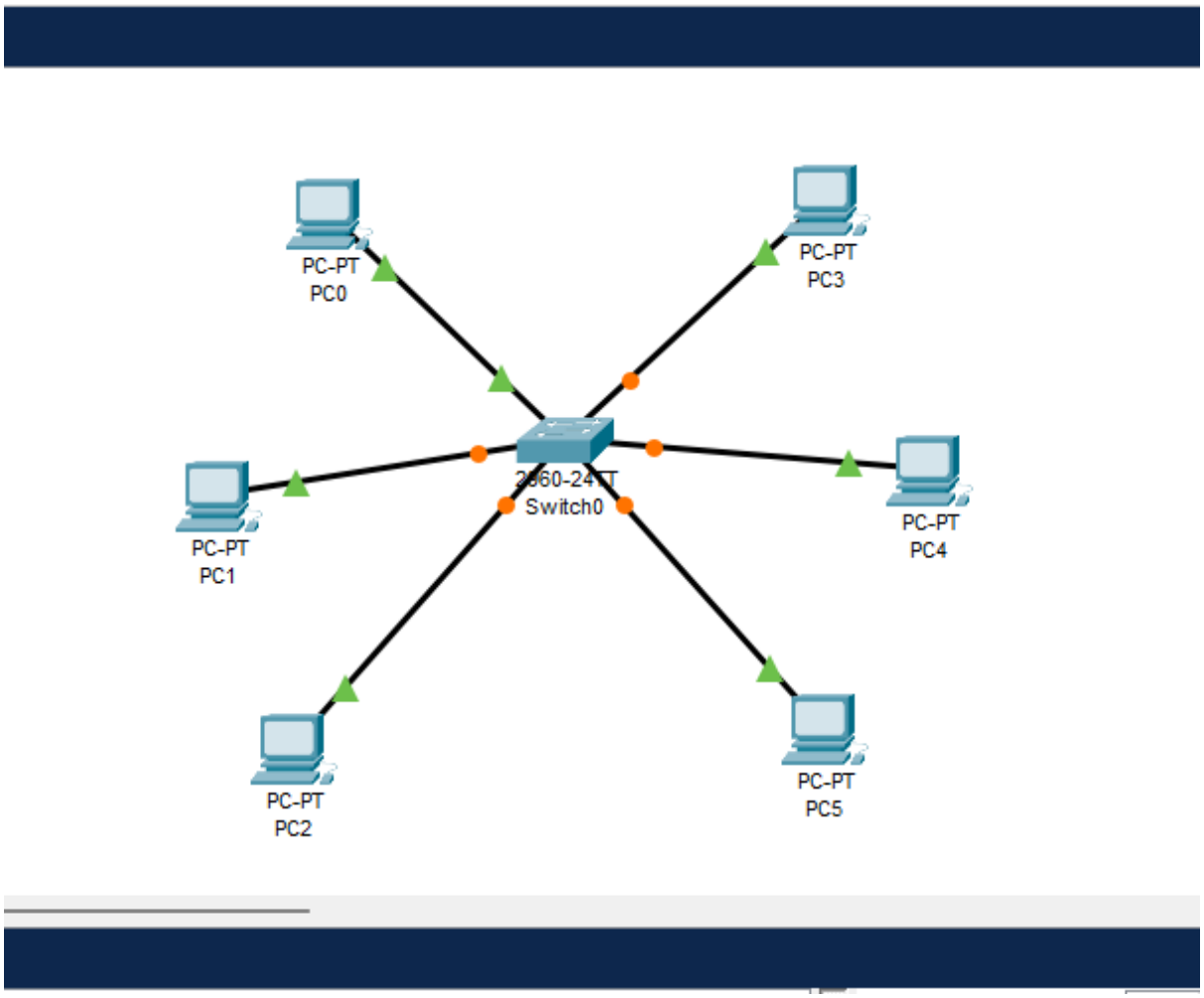
CSMA/CD (Switch-Based Network)

Procedure

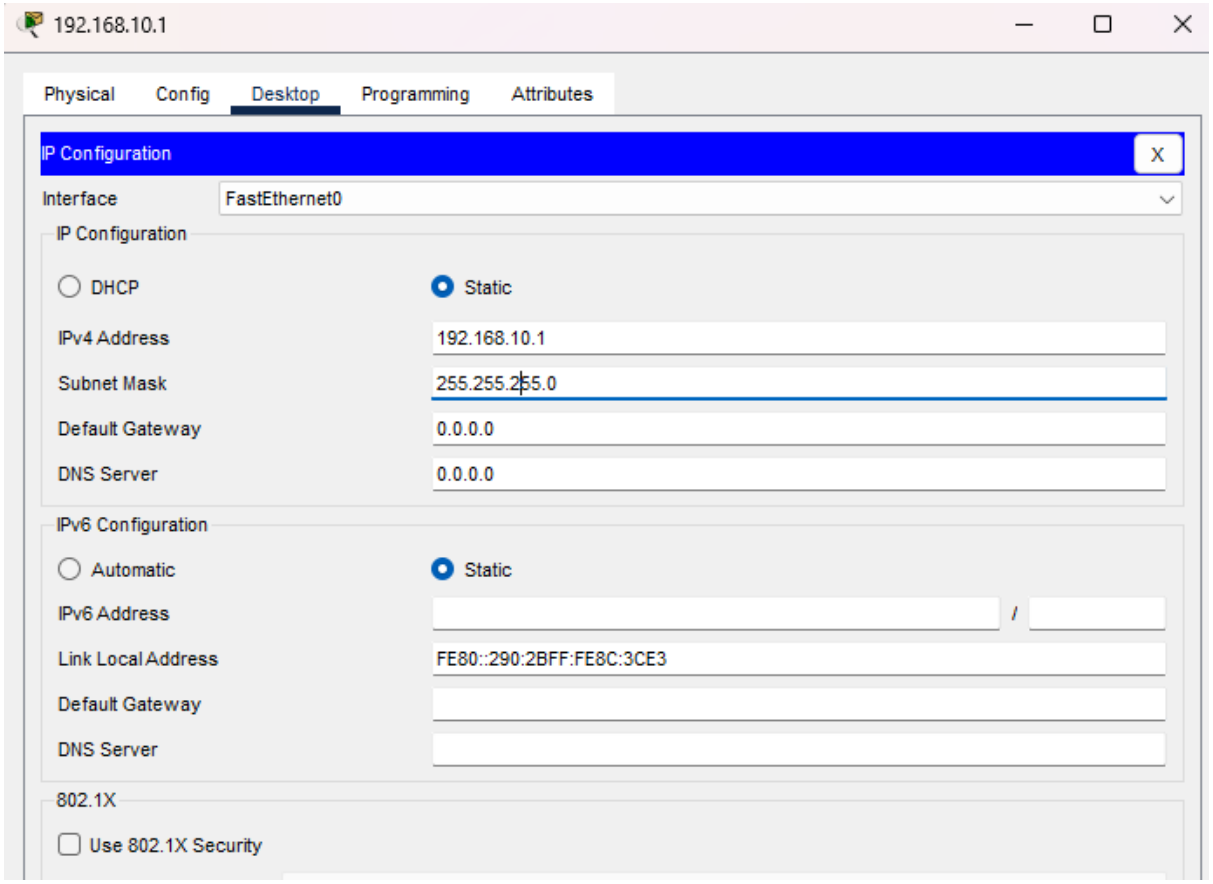
1. Open Cisco Packet Tracer and create a new project.
2. Add devices:
    - Drag and drop **4–6 PCs**
    - Add **1 Switch (2960)**

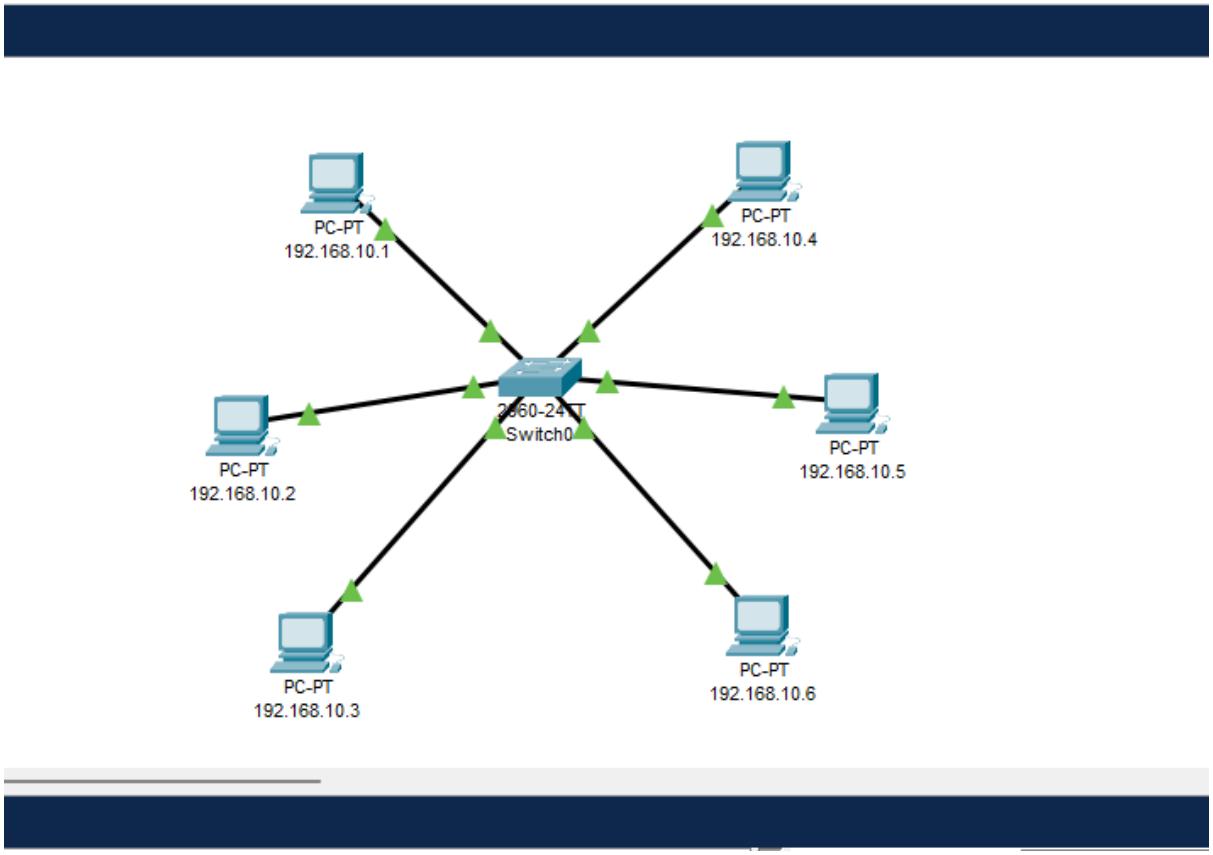


3. Connect devices:
   - Use **Copper Straight-Through Cable**
   - Connect all PCs to the switch

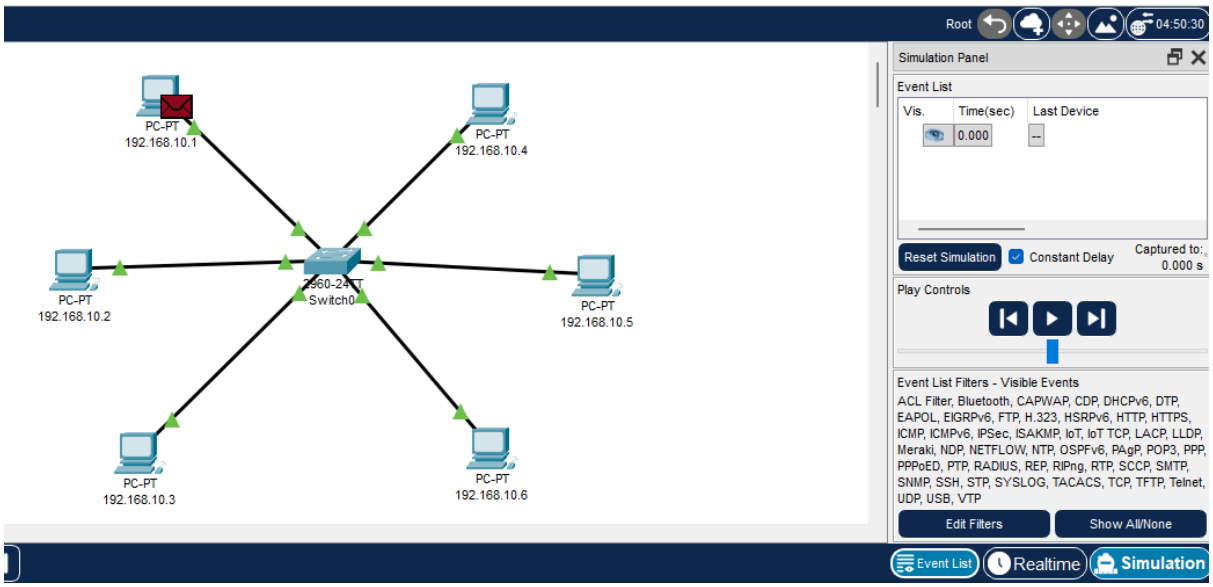


4. Configure IP addresses:
    - Assign IP addresses in the same network





5. Simulate data transmission:
    - Use **Add Simple PDU tool**
    - Send packets between different PCs



6. Enable Simulation Mode:
    - Switch to **Simulation Mode**
    - Click **Auto Capture/Play**
7. Observe:
    - Packet flow through switch

- Collision handling
- Data transmission efficiency